Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CE741]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

CONSTRUCTION PLANNING AND MANAGEMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. List and explain the step involved in planning process **05**

b. List and explain different stages of planning **09**

2. a. A project consists of the following activities draw the network diagram. Calculate the activity floats and the critical path? **08**

ACTIVITY	DURATION(Weeks)
1-2	10
1-3	20
3-4	8
4-6	10
3-5	12
6-7	8
5-7	6

b. Explain difference between CPM and PERT. **06**

3. Define the following terms **10**

- **i.** Direct cost
- **ii.** indirect cost
- **iii.** Total project cost
- **iv.** Normal cost
- **v.** Normal time
- **vi.** Crash Cost
- **vii.** Crash Time

OR

4. For taking up crashing of project duration determine **14**

- **i.** Direct cost-duration relationship05
- **ii.** Total cost-duration relationship
- **iii.** The optimum duration of the project against optimum total cost, for the information given as follow.

ACTIVITY	NORMAL DURATION (Days)	NORMAL COST (Rs)	CRASH DURATON (Days)	CRASH COST (Rs)
1-2	18	16000	12	19000
2-3	10	10000	6	11000

5. **a.** Explain the factors affecting labour output. **07**

b. Explain the classification of construction equipment. **07**

OR

6. **a.** What do you mean by resource levelling? List steps involved in resource leveling. **07**

b. What do you mean by resource smoothing? List steps involved in resource smoothing. **07**

7. **a.** Explain the organization for quality control with a flow chart. **07**

b. List and explain different elements of quality. **07**

OR

8. **a.** List and explain different types of concreting equipment. **07**

b. List and explain different earth moving equipment. **07**